

title: "Savitribai Phule Pune University --- Programming and Problem Solving"
code: "110005"
pattern: "2019"
total_marks: 70
duration: "2½ Hours"

Seat No.

SAVITRIBAI PHULE PUNE UNIVERSITY

Programming and Problem Solving (110005)

2019 Pattern | Semester I/II

Total Marks: 70 | Time: 2½ Hours

Instructions: Q1/Q2, Q3/Q4, Q5/Q6, Q7/Q8. Neat diagrams must be drawn wherever necessary.

Unit III --- Functions and Modules (18 Marks)

Q1) Attempt the following:

a) Define a function. Explain the need for functions in Python.

b) Differentiate between local and global variables with examples.

c) Write a program using a function to find the largest of 3 numbers by passing numbers as arguments.

OR

****Q2)** Attempt the following:**

**a) Explain the following types of function arguments with examples: i) Default arguments
ii) Variable length arguments**

b) Write a program to find the square of a number using lambda function.

c) What are modules and packages in Python? Explain a user-defined module with example.

Unit IV --- Strings (17 Marks)

Q3) Attempt the following:

a) Explain different ways of formatting a string in Python with suitable examples.

b) Explain different ways of iterating a string.

**c) What is the output of the following for string `S = "Learning Python is fun"`: i) `print(S[:10])`
ii) `print(S[::-1])` iii) `print("Is" not in S)` iv) `print(S[4])` v) `print(S[0:10])`**

OR

****Q4)** Attempt the following:**

**a) Explain the following string methods with examples: i) `isspace()` ii) `join()`
iii) `lower()`**

b) Write a program to display a string and count the number of characters in the string using a loop.

c) Explain the `ord()` and `chr()` functions with examples. Explain `in` and `not in` operators on strings.

Unit V --- Object Oriented Programming (18 Marks)

Q5) Attempt the following:

a) Explain the following programming paradigms: i) Monolithic Programming ii) Procedural Programming iii) Object Oriented Programming

- b) Explain the following OOP concepts with examples: i) **Class variable** ii) **Object variable** [5]
- c) Write a Python program to create a class `Employee` with attributes `Name`, `emp_id`, and `salary`. Display data of 4 employees. [4]
- OR
- **Q6)**** Attempt the following: [15]
- a) Explain the following features with examples: i) **Containership** ii) **Inheritance** [6]
iii) **Data Abstraction**
- b) Explain class method and self object with suitable example.** [5]
- c) Write a program to create a class `Car` with two attributes `name` and `cost`. Create two objects and display their information. [4]
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Unit VI --- File Handling (17 Marks)

- Q7) Attempt the following:** [15]
- a) **What is a file? Explain relative and absolute path of a file with examples.** [6]
- b) Explain the following file handling methods: i) `read(n)` ii) `readlines()` [5]
iii) `close()`
- c) **Explain the following file access modes: `r`, `w+`, `a+`, `rb`, `ab+`.** [4]
- OR
- **Q8)**** Attempt the following: [15]
- a) **Explain any three directory methods with examples.** [6]
- b) **Explain text and binary files with examples.** [5]
- c) **Explain any three dictionary methods with examples.** [4]
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*** END OF QUESTION PAPER ***